

Gravistat — Post-Approval Stability Protocol

Protocol STAB-P-2022-43

Sample Inventory and Pull Windows

Sample inventory per condition and timepoint is defined in the pull schedule appendix. Scheduled pulls are performed within ± 7 days of the nominal date; pull records are maintained in the site stability management system.

Analytical Procedures

Assay and degradation products are determined by the stability-indicating reversed-phase HPLC procedure described in the analytical methods dossier. Dissolution testing uses USP Apparatus II at 50 rpm. All procedures were validated in accordance with ICH Q2(R1) prior to protocol initiation.

Storage Conditions

The primary (refrigerated condition) storage condition for this protocol is $5^{\circ}\text{C} \pm 3^{\circ}\text{C}$. Accelerated studies at $40^{\circ}\text{C} \pm 2^{\circ}\text{C}$ / $75\% \text{ RH} \pm 5\% \text{ RH}$ were completed through six months during development and are not part of this protocol.

Testing Schedule

Long-term samples are tested at 0, 3, 6, 12, 18, 24, 36 and 48 months.

Stability Commitments

A post-approval commitment is made to enrol one commercial batch of Gravistat per year into the ongoing stability program and to report confirmed out-of-specification results within 30 days.

Stability Acceptance Criteria

The stability acceptance criterion applied at each interval is — Total degradation products: not more than 2.0%. All other registered specification tests apply unchanged.

Container Closure and Orientation

Samples are stored in the commercial container closure system. Bottles are stored upright; a subset of inverted samples is included at the initial and final timepoints to assess closure-liner interaction. Induction seal integrity is verified at each pull.

Batch Selection

The protocol covers three primary batches manufactured at commercial scale using the approved process. Batch genealogy, manufacturing dates, and packaging lots are recorded in the batch appendix. All batches were released against the approved specification prior to set-down.

Bracketing and Matrixing

Bracketing is justified under ICH Q1D on the basis that the 10 mg and 100 mg presentations bracket the 40 mg presentation with respect to fill volume and surface-to-volume ratio.

Deviations

Chamber excursion of +3°C for 12 hours affected the 24-month interval for batch 001; mean kinetic temperature remained within limits.

Container Closure and Orientation

Samples are stored in the commercial container closure system. Bottles are stored upright; a subset of inverted samples is included at the initial and final timepoints to assess closure-liner interaction. Induction seal integrity is verified at each pull.

Photostability

Confirmatory photostability testing per ICH Q1B was completed during development and is not repeated in this protocol. The product is labeled for storage in the original container to protect from light.

Data Evaluation and Reporting

Stability data are evaluated per ICH Q1E. Regression analysis is applied to quantitative attributes exhibiting change over time; poolability across batches is assessed at the 0.25 significance level. Annual reports summarize all data generated in the reporting interval.